

Submission STAT1-PS1-SUB08

Question STAT1-PS1-Q01

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visible work:

$$\bar{x} = 33/5 = 6.60 \dots ?$$

last line is hard to read; maybe mean=6.60; sample variance=18.30

Question STAT1-PS1-Q02

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visible work:

$$P(D_1 \cap D_2) = (3/16)((3-1)/(16-1)) \dots ?$$

last line is hard to read; maybe 0.0250

Question STAT1-PS1-Q03

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visible work:

$$SE = \sqrt{(p(1-p)/n)} = \sqrt{(0.35(1-0.35)/40)} = 0.0754 \dots ?$$

last line is hard to read; maybe (0.202, 0.498); repeated-sampling interpretation

Question STAT1-PS1-Q04

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visible work:

Increase sample size n to reduce standard error.... ?

last line is hard to read; maybe increase n; lower confidence level or variability; note the coverage trade-off

Question STAT1-PS1-Q05

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visible work:

$0.03 < 0.05$, so reject H_0 under the stated rule.... ?

last line is hard to read; maybe reject H_0 ; evidence against H_0 , not proof that H_0 is false

Question STAT1-PS1-Q06

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visible work:

$$\bar{x} = 21/5 = 4.20 \dots ?$$

last line is hard to read; maybe mean=4.20; sample variance=4.70